THE CONSTANTS OF THE FUNCTIONAL EQUATIONS OF L-FUNCTIONS

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Contents

Introduction

The essential results of this article are the following.

(A) We give a simple proof of Langlands' theorem [9], already proved up to sign by Dwork [6], which makes it possible to write the constant in the functional equations of Artin L-functions, or of Weil L-functions [19], as a product of local constants.

The theorem to be proved is local, and Langlands' proof had the elegance of being purely local. Essentially, the problem is to prove a multiplicative identity between somewhat generalized Gauss sums whenever one has a $\mathbb{Z}$-linear relation between characters of local Galois groups induced by characters of one-dimensional representations of subgroups. Langlands constructed a generating set of such relations, and reduced the identities to be proved to identities between Gauss sums, which he proved using Stickelberger's theorem.

The proof presented here is very close to the proof [5] of the Hasse–Davenport identity (cf. also Weil [18]). It uses a global argument (4.12), and the fact that the local constants to be defined are essentially trivial at infinity, in the unramified case, and in certain very ramified cases (4.14).

(B) Let K be a function field and

$$\rho : \mathrm{Gal}(\overline{K}/K) \longrightarrow \mathrm{GL}(n, \mathbb{Q}_\ell)$$

an ℓ-adic representation of the Galois group of K, unramified almost everywhere. Grothendieck showed that the Euler product over the places of K, with p^{-s} replaced by an indeterminate t,

$$Z(\rho, t) = \prod_v \det\bigl(1 - F_v t^{\deg(v)}, (\mathbb{Q}_\ell^n)^{\rho(I_v)}\bigr)^{-1} \in \mathbb{Q}_\ell[[t]],$$

is a rational function, and that a functional equation relates $Z(\rho, t)$ to

$$Z^\vee(\rho, t): \qquad Z^\vee(\rho, p^{-1}t^{-1}) = \varepsilon\, Z(\rho, t),$$

where the constant ε is a monomial at^b. This theory is summarized in §10. Result (A) suggests a formula expressing ε as a product of local constants. We prove that this formula is true when ρ belongs to an infinite compatible system of ℓ-adic representations (9.3 and 9.9). According to Weil [16] and Jacquet–Langlands [8], this formula, applied to the compatible system of ℓ-adic representations defined by an elliptic curve over K, makes it possible to attach to every elliptic curve over K a modular form for $\mathrm{GL}(2, K)$.

It would be very interesting to know whether the formula obtained remains valid for an arbitrary ℓ-adic representation.

Here is the content of the various sections, and their interdependence.

The beginning of §1 gathers a few results on induced representations of finite groups which will be useful to us.

The end of §1, from 1.11 to the end, is not used in the rest of the article. For a solvable group G, it describes a set R of $\mathbb{Z}$-linear relations between the characters of G induced by one-dimensional characters of subgroups, and proves that every such relation is a linear combination of elements of R. The result and its proof are implicitly contained in [9].

Sections 2 and 3 serve mainly to fix notation. In §2 we recall the structure of the Galois group of a local field [10], define the Weil groups, a variant of Galois groups, and give a few statements from class field theory. The reader should note that throughout this article the isomorphism of local class field theory sends a uniformizer to the inverse of the Frobenius substitution; the reason is given in 3.6. Section 3 recalls Tate's thesis and Artin [1].

Section 4 contains the proof of result (A) above. The formula sheet in §5 gives a list of fundamental properties of the local constants constructed in §4, and a few identities between Gauss sums that follow from them.

Sections 6, 7, and 8, the latter depending only on §2, are preliminaries to the proof of (B), given in §9. In the statements of §§3 and 4, we were very careful to use only rational formulas, involving neither passage to a limit nor square roots, even of a positive integer. We harvest the benefit of this in §§6 and 7. In §6, for K a non-archimedean local field and V a modular representation of $W(\overline{K}/K)$ over a field A of characteristic $\ell \neq p$, we define, by transport of structure for $\ell = 0$ and by reduction modulo ℓ for $\ell \neq 0$, a local constant

$$\varepsilon(V, \psi, dx) \in A^*.$$

In §7, for K global of characteristic p and V a modular representation of $W(\overline{K}/K)$, we prove, by reduction modulo ℓ, a functional equation for the corresponding L-function $L \in \Lambda(t)$.

In §8 we give the isomorphism classes of ℓ-adic representations of the Weil group of a local field a purely algebraic description, independent of the topology of $\mathbb{Q}_\ell$. This is used to define compatibility of ℓ-adic and ℓ'-adic representations. The notion introduced is finer than Serre's [12].

Section 9 gives the proof of (B) (9.3 and 9.9). The idea is that, to prove an identity, it is enough to prove it modulo ℓ for infinitely many ℓ. We also give, for function fields, a partial answer to a question of Serre by showing (9.8) that two ℓ- and ℓ'-adic representations giving the same characteristic polynomial of Frobenius, assumed to lie in $\mathbb{Q}[t]$, at almost all places satisfy compatibility at the residual places.

Section 10 summarizes Grothendieck's theory of L-functions (cf. [7] and SGA 5), which is used crucially in §9. In 10.11 I try to explain the point of his method. The odd results 10.13 follow immediately from what precedes. I do not know their meaning.

§1. Virtual representations of a finite group

1.1. Let K be a commutative field. The most important case for us will be that in which K is algebraically closed of characteristic 0, for example $K = \mathbb{C}$. Let G be a group. We shall write $R_K(G)$ for the Grothendieck group of the category of finite-dimensional $K[G]$-modules over K. It is identified with the free $\mathbb{Z}$-module with basis the set of isomorphism classes of irreducible K-linear representations of G. It is a commutative ring with unit, for tensor product over K, and even a λ-ring in the sense of Grothendieck. Its elements are the virtual representations of G over K.

To define a homomorphism f from $R_K(G)$ to a commutative group X, it is enough:

a) to define $f(V)$ for V a representation of G over K;

b) to check that for every exact sequence of representations

$$0 \longrightarrow V' \longrightarrow V \longrightarrow V'' \longrightarrow 0,$$

one has $f(V) = f(V') + f(V'')$.

Applying this rule, one defines the dimension of a virtual representation

$$\dim : R_K(G) \longrightarrow \mathbb{Z},$$

and its determinant

$$\det : R_K(G) \longrightarrow \mathrm{Hom}(G, K^*).$$

If H is a subgroup of finite index in G, define induction

$$\mathrm{Ind}_H^G : R_K(H) \longrightarrow R_K(G)$$

as corresponding to extension of scalars

$$V \longmapsto K[G] \otimes_{K[H]} V.$$

This is an exact functor, and therefore passes to virtual representations. For $f : G \to H$, define restriction

$$\mathrm{Res}_G^H : R_K(H) \longrightarrow R_K(G)$$

as restriction of scalars from $K[H]$ to $K[G]$. For f surjective one sometimes speaks rather of inflation.

For every group G, write G^{ab} for the largest abelian quotient of G. A K-quasi-character χ of G is a homomorphism

$$\chi : G \longrightarrow K^*;$$

we identify it with the K-quasi-character of G^{ab} through which it factors, and denote by $[\chi]$ the corresponding one-dimensional representation. For G finite, we shall use the words character and quasi-character interchangeably. The word character will never mean the character of a representation of dimension > 1.

The following proposition is due to P. Gallagher [20].

Proposition 1.2. Let G be a group, H a subgroup of finite index, and

$$t : G^{\mathrm{ab}} \longrightarrow H^{\mathrm{ab}}$$

the transfer. For ρ a virtual representation of dimension 0 of H, and $x \in G^{\mathrm{ab}}$, one has

$$\det(\mathrm{Ind}_H^G(\rho))(x) = \det(\rho)(t(x)).$$

We shall prove, in only slightly greater generality, that if ε is the determinant of the permutation representation of G on G/H,

$$\varepsilon : G \longrightarrow \mathfrak{S}_{G/H} \longrightarrow \{\pm 1\},$$

then, for ρ of arbitrary dimension,

$$\det(\mathrm{Ind}_H^G(\rho))(x) = \varepsilon^{\dim \rho} \det(\rho)(t(x)).$$

This is fairly evident in topological terms. Let BG be the classifying space of G, and let B_H be that of H, regarded as a covering of B_G. Representations of G are identified with local systems of finite-dimensional K-vector spaces on B_G, and Ind_H^G with the functor f_*. The transfer $H^1(B_G, X) \to H^1(B_H, X)$, for $X = K^*$, corresponds to the norm $f_!$. If $\rho \in H^1(B_G, K^*)$ is the class of a rank-one local system of K-vector spaces, then $f_!(\rho)$ is the class of $N(L)$, the local system on B_G which, locally on B_G, is the tensor product of the inverse images of L by the $[G : H]$ disjoint sections.

Put

$$\det V = \bigwedge^{\dim V} V$$

for a local system V, and let ε be the local system $\det f_* K$. Proposition 1.2 follows from the existence of a canonical isomorphism, for V a local system on B_H,

$$\det f_* V \simeq \varepsilon^{\otimes \dim V} \otimes N(\det V).$$

The construction of this isomorphism is local on B_G, and reduces to the following construction.

Construction 1.3. For $(V_i)_{i \in I}$ a finite family of K-vector spaces of dimension d, and $\varepsilon = \det(K^I)$, one has canonically

$$\det\Big(\bigoplus_{i \in I} V_i\Big) \simeq \varepsilon^{\otimes d} \otimes \bigotimes_{i \in I} \det(V_i).$$

For $i_1, \ldots, i_N$ the sequence of elements of I, the map is

$$\begin{aligned}&(e_{i_1,1} \wedge \cdots \wedge e_{i_1,d}) \wedge \cdots \wedge (e_{i_N,1} \wedge \cdots \wedge e_{i_N,d})\\ &\quad \longmapsto (i_1 \wedge \cdots \wedge i_N)^{\otimes d} \otimes (e_{i_1,1} \wedge \cdots \wedge e_{i_1,d}) \otimes \cdots \otimes (e_{i_N,1} \wedge \cdots \wedge e_{i_N,d}).\end{aligned}$$

Parenthesis 1.4. Instead of a classifying space BG, one could more intrinsically have used, in the preceding argument, Grothendieck's classifying topos B_G (SGA 4, IV, 2.4, p. 315, for $E = (\mathrm{Ens})$).

Assume G finite. We shall say that K is large enough if it contains all roots of unity whose orders divide the l.c.m. of the orders of elements of G. A group H will be called p-elementary if it is the product of a p-group and a cyclic group; elementary if it is p-elementary for some prime p. We shall need the following variant of Brauer's theorem on induced characters, obtained by omitting everywhere "of dimension zero."

Proposition 1.5. If K is large enough, every virtual representation of dimension 0 of G is a sum of virtual representations

$$\mathrm{Ind}_H^G(\chi),$$

where χ is virtual of dimension 0, H is elementary, and χ is a linear combination of one-dimensional representations of H.

By Brauer's theorem applied to the trivial representation 1_G of G, there is a decomposition

$$1_G = \sum_H \mathrm{Ind}_H^G(z_H), \qquad H \text{ elementary}.$$

For y a virtual representation of dimension 0 of G, one then has

$$y = \sum_H y \cdot \mathrm{Ind}_H^G(z_H) = \sum_H \mathrm{Ind}_H^G(\mathrm{Res}_H^G(y) \cdot z_H),$$

with $\dim(\mathrm{Res}_H^G(y) \cdot z_H) = 0$. This reduces us to the case where G is elementary.

By linearity, one may suppose that y is of the form

$$\rho - \dim(\rho)\, 1_G,$$

with ρ irreducible, hence, since G is nilpotent, induced by a character χ of a subgroup H of G containing the center Z. Then

$$\rho - \dim(\rho)\, 1_G = \mathrm{Ind}_H^G(\chi - 1_H) + \big(\mathrm{Ind}_H^G(1_H) - [G : H] 1_G\big).$$

The first term is of the desired type, and the second is the inflation of a virtual representation of dimension 0 of G/Z. One concludes by induction on the order of G, noting that $|G/Z| < |G|$ if $G \neq \{e\}$.

Scholium 1.6. A homomorphism

$$f : R_K(G) \longrightarrow X$$

is known once its value on 1_G and on the

$$\mathrm{Ind}_H^G(\chi - 1_H),$$

for H elementary and χ a character of H, are known.

1.7. Let A be a discrete valuation ring of unequal characteristic p, with fraction field K and residue field $k = A/\mathfrak{m}$. Recall the definition of the decomposition homomorphism ([11], III, 2.2)

$$d : R_K(G) \longrightarrow R_k(G) :$$

for V a representation of G over K, and E a G-invariant lattice,

$$d([V]) = [E \otimes_A k].$$

Proposition 1.8. Assume that K is sufficiently large. The kernel of the decomposition homomorphism d is then generated by the elements

$$\mathrm{Ind}_H^G(\chi') - \mathrm{Ind}_H^G(\chi'')$$

where H is an elementary subgroup and $\chi', \chi'' \in \mathrm{Hom}(H, K^*)$ are congruent modulo $\mathfrak{m}$.

Since the ring homomorphism d commutes with induction, the argument of 1.5 reduces us to the case where G is elementary, hence the product of a p-group P and a group H of order prime to p. Since K and k are sufficiently large, one has

$$R(G) = R(P) \otimes R(H),$$

and similarly over k. For a group of order prime to p, d is bijective. Applying Brauer's theorem to H, it is enough to prove 1.8 for a p-group. This case is trivial, since every character $\chi \in \mathrm{Hom}(P, K^*)$ of a p-group is congruent to 1 modulo $\mathfrak{m}$.

1.9. For the rest of this subsection, assume that $K = \mathbb{C}$, consider only finite groups, and abbreviate $R_{\mathbb{C}}(G)$ to $R(G)$. For G commutative, write G^* for the group of characters of G.

Let $R^+(G)$ be the free $\mathbb{Z}$-module with basis the set of G-conjugacy classes of pairs (H, χ), where H is a subgroup of G and χ a character of H. Define a multiplication in $R^+(G)$ by

$$(A, \alpha)\,(B, \beta) = \sum_{(x,y) \in G \backslash (G/A \times G/B)} \big(A^x \cap B^y,\ (\alpha^x|_{A^x \cap B^y})(\beta^y|_{A^x \cap B^y})\big), \tag{1.9.1}$$

where (x, y) runs through representatives of the orbits of G in $G/A \times G/B$, $A^x = xAx^{-1}$, and α^x is the character $a \mapsto \alpha(x^{-1}ax)$ of A^x. With this multiplication, $R^+(G)$ is a commutative ring with unit $(G, 1)$, and one defines a ring homomorphism

$$b : R^+(G) \longrightarrow R(G), \qquad (H, \chi) \longmapsto \mathrm{Ind}_H^G(\chi). \tag{1.9.2}$$

This homomorphism is surjective by Brauer's theorem. For χ a character of G, multiplication by (G,χ) is denoted $\cdot\chi$ and is called twisting by χ:

$$(H,\varphi)\cdot\chi = (H,\varphi\cdot\chi|_H). \tag{1.9.3}$$

For $G' \subset G$, define induction

$$\mathrm{Ind}_{G'}^{G} : R^+(G') \longrightarrow R^+(G), \qquad (H,\chi) \longmapsto (H,\chi). \tag{1.9.4}$$

For a morphism $u : G \to G'$, define restriction by

$$\mathrm{Res}_u(H,\chi) = \sum_{s\in u(G)\backslash G'/H} \big(u^{-1}(sHs^{-1}),\, \chi^s \circ u\big), \tag{1.9.5}$$

where s runs over a set of representatives of the double classes. There are commutative diagrams

$$\begin{array}{ccc} R^+(G') & \xrightarrow{\mathrm{Ind}} & R^+(G) \\ \downarrow{\scriptstyle b} & & \downarrow{\scriptstyle b} \\ R(G') & \xrightarrow[\mathrm{Ind}]{} & R(G) \end{array} \qquad \begin{array}{ccc} R^+(G') & \xrightarrow{\mathrm{Res}} & R^+(G) \\ \downarrow{\scriptstyle b} & & \downarrow{\scriptstyle b} \\ R(G') & \xrightarrow[\mathrm{Res}]{} & R(G), \end{array}$$

for the second, see [11], p. 75. For $R \in R^+(G)$ one has

$$R\cdot(H,\chi) = \mathrm{Ind}_H^G\big(\mathrm{Res}_H^G(R)\cdot\chi\big). \tag{1.9.6}$$

When u is surjective, one speaks of inflation.

Notation 1.9.8. For every additive function of representations F, write the same symbol for the function it defines on $R(G)$, and for $F \circ b$. Thus

$$\dim(v) = \dim(b(v)), \quad \ldots$$

Variants 1.10. (i) Let $R_0^+(G)$ be the subgroup of $R^+(G)$ generated by the $(H,\chi)-(H,1)$. As (H,χ) runs through the canonical basis of $R^+(G)$, except for the $(H,1)$, these elements form a basis of $R_0^+(G)$. One checks that $R_0^+(G)$ is an ideal of $R^+(G)$. Let $R^0(G) \subset R(G)$ be the set of virtual representations of dimension 0 of G. By 1.5, the natural map, induced by (1.9.2), from $R_0^+(G)$ to $R^0(G)$ is surjective.

(ii) For a topological group G, finite or not, $R^+(G)$ and $R_0^+(G)$ denote the analogous groups obtained by restricting to closed subgroups H of finite index and continuous quasi-characters. The functorialities of 1.9 generalize mutatis mutandis.

The rest of this subsection will not be used below. We give results drawn from Langlands [9].

Lemma 1.11. Suppose that $G = H\cdot C$, where C is commutative and normal, and let χ be a character of H. For every $\mu \in C^*$, let $G_\mu \subset G$ be the stabilizer of μ under conjugation. If

$$\chi|_{H\cap C} = \mu|_{H\cap C},$$

write $\{\chi,\mu\}$ for the character of $G_\mu = (H\cap G_\mu)\cdot C$ which extends $\chi|_{H\cap G_\mu}$ and μ. Then

$$\mathrm{Ind}_H^G(\chi) = \sum_{\substack{\mu\in C^*/G \\ \chi|_{H\cap C}=\mu|_{H\cap C}}} \mathrm{Ind}_{G_\mu}^G\{\chi,\mu\}.$$

The verification is left to the reader.

1.12. An element R of $\mathrm{Ker}(R^+(G) \to R(G))$ will be said to be of type I, II, or III if there exists a quotient A of a subgroup B of G such that R is obtained from an element S of one of the following respective types in $\mathrm{Ker}(R^+(A) \to R(A))$, by inflation from A to B, twisting by a character of B, and induction from B to G. Let ℓ be a prime number.

I. $A \simeq \mathbb{Z}/\ell\mathbb{Z}$, and

$$S = (e, 1) - \sum_{\chi \in A^*} (A, \chi).$$

II. A is a central extension of $(\mathbb{Z}/\ell\mathbb{Z})^2$ by an abelian group Z. Let H_i, $i = 1, 2$, be the inverse images in A of the first and second factors $\mathbb{Z}/\ell\mathbb{Z}$, and let χ_i be a character of H_i. Assume that $\chi_1|_Z = \chi_2|_Z$, and that this character of Z is nontrivial on a commutator. The relation S is

$$S = (H_1, \chi_1) - (H_2, \chi_2).$$

III. A is a semidirect product $H \cdot C$, χ is a character of H, and the normal subgroup C is minimal among the nontrivial commutative normal subgroups of A. For every character μ of C, A_μ is the stabilizer of μ, and $\{\chi, \mu\}$ is the character of A_μ extending χ and μ. The relation S expresses

$$S = (H, \chi) - \sum_{\mu \in C^*/A} (A_\mu, \{\chi, \mu\}),$$

where μ runs over a set of representatives for the orbits of A in C^*.

These relations are special cases of 1.11: for I, $H = \{e\}$ and $C = A$; for II, $H = H_1$ and $C = H_2$; for III, H and C are as indicated.

Every relation obtained by induction, inflation, or twisting from a relation of type I, respectively II or III, is again of the same type.

Theorem 1.13 (Langlands [9], §18). If G is nilpotent, the kernel of

$$b : R^+(G) \longrightarrow R(G)$$

is generated, as a $\mathbb{Z}$-module, by the relations of type I and II.

Lemma 1.13.1. If G is commutative, $\mathrm{Ker}(b)$ is generated by the relations of type I.

One has to show that, for every subgroup H of G and every $\chi \in H^*$, the relation

$$(H, \chi) - \sum_{\psi \in (G/H)^*} (G, \tilde{\chi}\psi)$$

is a consequence of relations of type I. These are the following: for $H' \subset H'' \subset G$, with $[H'' : H']$ prime, and χ a character of H', always induced by a character of G,

$$\mathrm{Ind}_{H'}^{G}(\chi) - \sum_{\chi''|_{H'} = \chi} \mathrm{Ind}_{H''}^{G}(\chi'') = 0.$$

The general relation follows by taking a Jordan–Hölder series of G/H.

Lemma 1.13.2. Let C be a commutative normal subgroup of G, let T be a set of representatives of C^*/G, let $\mu \in T$, and let G_μ be the stabilizer of μ. Let (H_i) be a family of subgroups of G containing C, and let χ_i be a character of H_i, with $\chi_i|_C \in T$. If the relation

$$\sum_i n_i \, \mathrm{Ind}_{H_i}^{G}(\chi_i) = 0$$

holds in $R(G)$, then

$$\sum_{\chi_i|_C = \mu} n_i \, \mathrm{Ind}_{H_i}^{G_\mu}(\chi_i) = 0$$

holds in $R(G_\mu)$. The verification is left to the reader.

Let Z be the center of G, and prove 1.13 by induction on the order of G/Z. By 1.13.1, we may suppose G noncommutative. Let C be a commutative normal subgroup containing Z, with $[C : Z] = \ell$ prime, and with central image in G/Z.

Consider a relation

$$\sum_i n_i \operatorname{Ind}_{H_i}^G(\chi_i) = 0. \tag{1}$$

It is a consequence of relations induced by

$$\sum_{\chi_i|_{H_i\cap Z}=\chi} \operatorname{Ind}_{H_i}^G(\chi_i) = \operatorname{Ind}_{H_iZ}^G(\chi)$$

and of a relation (2) analogous to (1), with $H_i \supset Z$. The displayed relations are linear combinations of relations of type I; this follows from 1.13.1 applied to $H_iZ/\operatorname{Ker}(\chi_i)$.

The relation (2) is a consequence of relations induced from 1.11 for HC, H, C, χ, with $H \supset Z$,

$$\operatorname{Ind}_H^{HC}(\chi) = \sum_{\substack{\mu\in C^*/H \\ \chi|_{H\cap C}=\mu|_{H\cap C}}} \operatorname{Ind}_{G_\mu}^{HC}\{\chi,\mu\}, \tag{b}$$

and of a relation (3) analogous to (1), this time with $H_i \supset C$. Apply 1.13.2 to (3), to express it as a sum of relations induced by relations

$$\sum_i n_i \operatorname{Ind}_{H_i}^{G_\mu}(\chi_i) = 0.$$

The image of C in $G/\operatorname{Ker}(\mu)$ is central, and the induction hypothesis therefore applies to G_μ; the resulting relations are of the desired type.

It remains to prove that the relations (b) are of the desired type. If $HC \neq G$, this follows from the induction hypothesis applied to HC. If $HC = G$, then H is normal, because it contains Z and C is central modulo Z. Let A be the intersection of the kernels of the conjugates of χ. The relation (b) is the inflation of an analogous relation for G/A. If the induction hypothesis applies to G/A, we are done. Otherwise we are in the same situation as before, with $A = \{e\}$, and we conclude by the following lemma.

Lemma 1.13.3. With the preceding notation, if $A = \{e\}$, the relation (b) is of type II.

The group H is commutative, since characters separate its elements. The group Z is cyclic, since one character separates its elements. For $c \in C$ and $h \in H$, put

$$u(c,h) = chc^{-1}h^{-1}.$$

This is an element of Z, since C is central modulo Z. It depends biadditively on c and h, and defines

$$u : C/Z \otimes H/Z \longrightarrow Z.$$

Let c generate C/Z. Since Z is the center and $G \neq Z$, the map $u(c,\ ) : H/Z \to Z$ is injective, with nontrivial cyclic image killed by ℓ. Hence $|C/Z| = |H/Z| = \ell$, and G is a central extension of $C/Z \times H/Z$ by Z. The character χ is distinct from at least one of its conjugates, otherwise H would be central; it is therefore nontrivial on a commutator, and we are in situation II.

Theorem 1.14 (Langlands [9]). If G is solvable, the kernel of

$$b : R^+(G) \longrightarrow R(G)$$

is generated by the relations of type I, II, and III.

Lemma 1.14.1. If G is solvable, the $\mathbb{Z}$-submodule $N(G)$ of $R^+(G)$ generated by the relations of type I, II, and III is an ideal.

We prove 1.14 and 1.14.1 by simultaneous induction on the order of G.

(A) Theorem 1.14 for the proper subgroups of G implies 1.14.1 for G. Let $R \in R^+(G)$ be of type I, II, or III, let H be a subgroup, and let χ be a character of H. We prove that

$$R \cdot (H, \chi) \in N(G).$$

If $H = G$, this follows from the stability of $N(G)$ under twisting by a character of G. If $H \neq G$, then by (1.9.6)

$$R \cdot (H, \chi) = \mathrm{Ind}_H^G(\mathrm{Res}_H^G(R) \cdot \chi).$$

By the hypothesis, $\mathrm{Res}_H^G(R) \in N(H)$, and twisting followed by induction gives an element of $N(G)$.

(B) Lemma 1.14.1 for G, together with Theorem 1.14 for groups of smaller order, implies 1.14 for G. Let Z be the center of G. By 1.13, we may and shall suppose that G is not nilpotent. There are two cases.

(B1) $Z \neq \{e\}$. By Brauer's theorem in G/Z, there exist nilpotent subgroups $H_i \supset Z$ of G, and characters χ_i of H_i/Z, such that

$$\sum_i n_i \ \mathrm{Ind}_{H_i/Z}^{G/Z}(\chi_i) = 1_{G/Z}.$$

Since this relation lifts to a relation S in $R^+(G)$, and since the induction hypothesis applies to G/Z, one has $S - (G, 1) \in N(G)$. For $R \in \mathrm{Ker}(b)$, one has $R \cdot S \in N(G)$ by 1.14.1, while (1.9.6) reduces the terms to restrictions to the H_i, where the induction hypothesis applies. This proves the assertion.

(B2) $Z = \{e\}$. Let C be a nontrivial minimal commutative normal subgroup and, for $\mu \in C^*$, let G_μ be its stabilizer. Proceed as in the proof of 1.13. One finds that every relation $R \in \mathrm{Ker}(b)$ is a linear combination of:

(a) relations induced from 1.11 for HC, H, C, χ, with $H \subset G$, χ a character of H, and $H \not\supset C$;

(b) relations induced from identities in G_μ.

Since $Z = \{e\}$, one has $|G_\mu / \mathrm{Ker}(\mu)| < |G|$, and the induction hypothesis applies to the latter relations. Consider a relation of type (a). If $HC \neq G$, apply the induction hypothesis. If $HC = G$, then $H \cap C$ is normal, since it is normal in both H and C. By minimality of C, $H \cap C = \{e\}$, and the relation is of type III.

Remark 1.15. It is probably possible to extract from [9] a description of a system of generators of

$$\mathrm{Ker}(R_0^+(G) \longrightarrow R^0(G))$$

for suitable groups G. I have not had the courage to try.

2. Weil groups

2.1. Roots of unity. Let K be a separably closed field of characteristic exponent p. For every integer n, write $\mu_n(1)(K)$, or simply $\mu_n(1)$, for the group of n-th roots of unity in K. For $n \mid m$, there is a transition map

$$\mu_m(1) \longrightarrow \mu_n(1), \qquad x \longmapsto x^{m/n};$$

put

$$\widehat{\mathbb{Z}}(1) = \varprojlim_n \mu_n(1).$$

For a prime number ℓ, similarly put

$$\mathbb{Z}_\ell(1) = \varprojlim_r \mu_{\ell^r}(1).$$

The evident maps $\widehat{\mathbb{Z}}(1) \to \mathbb{Z}_\ell(1)$ induce an isomorphism

$$\widehat{\mathbb{Z}}(1) \xrightarrow{\sim} \prod_{\ell \neq p} \mathbb{Z}_\ell(1).$$

For every $\widehat{\mathbb{Z}}$-module V, put $V(1) = V \otimes_{\widehat{\mathbb{Z}}} \widehat{\mathbb{Z}}(1)$. For a $\mathbb{Z}_\ell$-module V, one has

$$V(1) \simeq V \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(1).$$

For $K = \mathbb{C}$, the group $\mu_n(1)$ is canonically isomorphic to the group of roots of unity in $\mathbb{C}$. For $x \in \mu_n(1)$ and $\xi \in \mathbb{Z}/n\mathbb{Z}$, the image of $\xi \otimes x$ in $\mathbb{Q}/\mathbb{Z}(1)$ is ξx, giving the usual identifications.

For $\ell \neq p$, $\mathbb{Z}_\ell(1)$ is a free rank-one $\mathbb{Z}_\ell$-module. For $i \in \mathbb{Z}$, write $\mathbb{Z}_\ell(i)$ for its i-th tensor power, and for $i < 0$ for the dual of $\mathbb{Z}_\ell(-i)$. For every $\mathbb{Z}_\ell$-module V, put again $V(i) = V \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(i)$.

2.2. Notation. Let K be a local field. We write $\| \; \|$ for the normalized absolute value of K. If dx is a Haar measure, then

$$\int f(ax)\,dx = \|a\|^{-1} \int f(x)\,dx,$$

i.e. $d(ax) = \|a\|\,dx$.

For $s \in \mathbb{C}$, write ω_s for the quasi-character

$$x \longmapsto \|x\|^s$$

from K^* to $\mathbb{C}^*$.

If K is nonarchimedean, i.e. $K \neq \mathbb{R}, \mathbb{C}$, write $\mathcal{O}$ for the valuation ring of the valuation v of K, π for a uniformizer, k for the residue field $\mathcal{O}/(\pi)$, p for the characteristic of k, and put

$$d = [k : \mathbb{F}_p], \qquad q = p^d = \#k.$$

Thus

$$\|x\| = q^{-v(x)}.$$

2.2.2. Suppose K is nonarchimedean. Let $\overline{K}$ be a separable closure of K, write $\overline{\mathcal{O}}$ for the integral closure of $\mathcal{O}$ in $\overline{K}$, let $\overline{k}$ be the residue field of $\overline{\mathcal{O}}$, an algebraic closure of k, and let v be the valuation of $\overline{K}$, with values in $\mathbb{Q}$, extending that of K.

There is a three-step filtration of the Galois group

$$\mathrm{Gal}(\overline{K}/K) \supset I \supset P$$

whose successive quotients are

$$\begin{array}{rcl} \mathrm{Gal}(\overline{K}/K)/I & \xrightarrow{\sim} & \mathrm{Gal}(\overline{k}/k) \xrightarrow{\sim} \widehat{\mathbb{Z}}, \\ I/P & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1)(\overline{k}), \\ P & : & \text{a pro-}p\text{-group.} \end{array}$$

It is obtained as follows; for proofs we refer to [10].

a) By transport of structure, $\mathrm{Gal}(\overline{K}/K)$ acts on $\overline{\mathcal{O}}$ and on $\overline{k}$. The inertia group I is the kernel of the reduction map

$$v' : \mathrm{Gal}(\overline{K}/K) \longrightarrow \mathrm{Gal}(\overline{k}/k).$$

The quotient $\mathrm{Gal}(\overline{k}/k)$ is canonically isomorphic to $\widehat{\mathbb{Z}}$, with topological generator the Frobenius substitution

$$\varphi : x \longmapsto x^q.$$

b) The wild inertia group P is the unique pro-p-Sylow subgroup of I. The equivariant morphism

$$t : I \longrightarrow \widehat{\mathbb{Z}}(1)(\overline{k}),$$

with kernel P, is characterized by the congruence, modulo the maximal ideal of $\overline{\mathcal{O}}$,

$$\frac{\sigma(x)}{x} = t(\sigma)\,v(x) \qquad (\sigma \in I,\ x \in \overline{K}).$$

For a prime $\ell \neq p$, write t_ℓ for the composite

$$I \longrightarrow \widehat{\mathbb{Z}}(1)(\overline{k}) \longrightarrow \mathbb{Z}_\ell(1)(\overline{k}).$$

2.2.3. Let k be a finite field with q elements, and let $\overline{k}$ be its algebraic closure. Let $W(\overline{k}/k)$ be the subgroup of $\mathrm{Gal}(\overline{k}/k) \simeq \widehat{\mathbb{Z}}$ generated by

$$\varphi : x \longmapsto x^q.$$

There is a canonical isomorphism

$$W(\overline{k}/k) \simeq \mathbb{Z}$$

with generator φ. The geometric Frobenius $F \in W(\overline{k}/k)$ is, by definition, φ^{-1}.

2.2.4. Let $K, \overline{K}$ be as in 2.2.2. The Weil group $W(\overline{K}/K)$, or absolute Weil group of K, is the subgroup of $\mathrm{Gal}(\overline{K}/K)$ consisting of the σ such that $v'(\sigma)$ is an integral power of Frobenius φ. It is endowed with the topology inducing the natural topology on I, and for which I is open. The group $\mathrm{Gal}(\overline{K}/K)$ is the completion of $W(\overline{K}/K)$ for the topology of open subgroups of finite index; if L is a finite extension of K in $\overline{K}$, the corresponding subgroup of $W(\overline{K}/K)$ is identified with $W(\overline{K}/L)$. We shall sometimes call geometric Frobenius elements those elements of $\mathrm{Gal}(\overline{K}/K)$ or $W(\overline{K}/K)$ whose image is F.

2.2.5. Suppose K is archimedean, with algebraic closure $\overline{K}$. Define $W(\overline{K}/K)$ as the following extension of $\mathrm{Gal}(\overline{K}/K)$ by $\overline{K}^*$:

1) $K \simeq \mathbb{R}$: $W(\overline{K}/K)$ is generated by $\overline{K}^*$ and an element F, with
$$F^2 = -1, \qquad FzF^{-1} = \overline{z} \quad (z \in \overline{K}^*).$$

2) $K \simeq \mathbb{C}$: $W(\overline{K}/K) = \overline{K}^*$.

2.3. Local class field theory provides an isomorphism between $W(\overline{K}/K)^{\mathrm{ab}}$ and K^*.

Assume K nonarchimedean. For a reason explained in 3.6, we normalize this isomorphism, i.e. choose it or its inverse, so that geometric Frobenius elements correspond to uniformizers:

$$\begin{tikzcd}
P \arrow[r, hook] \arrow[d] & I \arrow[r, hook] \arrow[d] & W(\overline{K}/K) \arrow[r] \arrow[d] & W(\overline{k}/k) \simeq \mathbb{Z} \arrow[d, "-1"] \\
1+(\pi) \arrow[r, hook] & \mathcal{O}^* \arrow[r, hook] & K^* \arrow[r, "v"] & \mathbb{Z}.
\end{tikzcd}$$

A representation of $W(\overline{K}/K)$ is unramified, respectively tamely ramified, if it is trivial on I, respectively on P. Similarly, a quasi-character χ of K^* is unramified, respectively tamely ramified, if it is trivial on $\mathcal{O}^*$, respectively on $1 + (\pi)$. For ℓ prime to p, the action of $\mathrm{Gal}(\overline{K}/K)$ on $\mathbb{Z}_\ell(1)$ is unramified, described by the quasi-character
$$\omega_1 = \|x\|$$
of K^*.

For K complex, the isomorphism of local class field theory is the evident one on 2.2.5; for K real, it is the one which makes 2.3.1 and 2.3.2 below true.

Let $L \subset \overline{K}$ be a finite extension of K, so that $W(\overline{K}/L) \subset W(\overline{K}/K)$. The following diagrams, where N is the norm and t the transfer, are commutative ([10], XIII, §4):

$$\begin{tikzcd}
W(\overline{K}/L) \arrow[r] \arrow[d] & W(\overline{K}/L)^{\mathrm{ab}} \arrow[r, "\sim"] \arrow[d] & L^* \arrow[d, "N_{L/K}"] \\
W(\overline{K}/K) \arrow[r] & W(\overline{K}/K)^{\mathrm{ab}} \arrow[r, "\sim"] & K^*,
\end{tikzcd} \tag{2.3.1}$$

$$\begin{tikzcd}
W(\overline{K}/K)^{\mathrm{ab}} \arrow[r, "\sim"] \arrow[d, "t"'] & K^* \arrow[d] \\
W(\overline{K}/L)^{\mathrm{ab}} \arrow[r, "\sim"] & L^*.
\end{tikzcd} \tag{2.3.2}$$

2.4. Let K be a function field in one variable over $\mathbb{F}_p$, and let k be its field of constants, the algebraic closure of $\mathbb{F}_p$ in K. Let $\overline{K}$ be a separable closure of K, let $\overline{k}$ be the algebraic closure of k in $\overline{K}$, and let $\mathrm{Gal}^0(\overline{K}/K)$ be the kernel of the restriction map from $\mathrm{Gal}(\overline{K}/K)$ to $\mathrm{Gal}(\overline{k}/k)$. The absolute global Weil group $W(\overline{K}/K)$ is, by analogy with 2.2.4, defined by the diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Gal}^0(\overline{K}/K) \arrow[r] \arrow[d, equal] & W(\overline{K}/K) \arrow[r] \arrow[d, hook] & W(\overline{k}/k) \arrow[r] \arrow[d, hook] & 0 \\
0 \arrow[r] & \mathrm{Gal}^0(\overline{K}/K) \arrow[r] & \mathrm{Gal}(\overline{K}/K) \arrow[r] & \mathrm{Gal}(\overline{k}/k) \arrow[r] & 0.
\end{tikzcd}$$

Let v be a place of K, and suppose a place of $\overline{K}$ above v has been chosen. There is then a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & I_v \arrow[r] \arrow[d, hook] & W(\overline{K}_v/K_v) \arrow[r] \arrow[d, hook] & W(\overline{k}_v/k_v) \arrow[r] \arrow[d, hook] & 0 \\
0 \arrow[r] & \mathrm{Gal}^0(\overline{K}/K) \arrow[r] & W(\overline{K}/K) \arrow[r] & W(\overline{k}/k) \arrow[r] & 0.
\end{tikzcd}$$

Global class field theory provides an isomorphism

$$W(\overline{K}/K)^{\mathrm{ab}} \xrightarrow{\sim} \mathbb{A}_K^*/K^*$$

where $\mathbb{A}_K^*/K^*$ is the idèle-class group, making the diagrams

$$\begin{tikzcd}
W(\overline{K}_v/K_v)^{\mathrm{ab}} \arrow[r] \arrow[d] & W(\overline{K}/K)^{\mathrm{ab}} \arrow[d] \\
K_v^* \arrow[r] & \mathbb{A}_K^*/K^*
\end{tikzcd}$$

commutative.

2.5. For the definition of global Weil groups in the case of number fields, we refer to Weil [19] or to [2], XIV. We shall use them only incidentally.

3. Reminders on L-functions

3.1. Let K be a local field and use again the notation of 2.2.1. We denote by dx a Haar measure on K, by d^*x a Haar measure on K^*, and by ψ a nontrivial additive character of K.

3.2. For a quasi-character, i.e. a continuous homomorphism,

$$\chi : K^* \longrightarrow \mathbb{C}^*,$$

define $L(\chi) \in \mathbb{C} \cup \{\infty\}$ as follows.

$$K \simeq \mathbb{R}. \tag{3.2.1}$$

Put

$$\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2),$$

and, for x the embedding of K in $\mathbb{C}$ and $N = 0$ or 1,

$$L(x^{-N}\omega_s) = \Gamma_{\mathbb{R}}(s).$$

$$K \simeq \mathbb{C}. \tag{3.2.2}$$

Put

$$\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s),$$

and, for z an embedding of K in $\mathbb{C}$ and $N \geq 0$,

$$L(z^{-N}\omega_s) = \Gamma_{\mathbb{C}}(s).$$

$$K \text{ nonarchimedean.} \tag{3.2.3}$$

Put

$$L(\chi) = 1 \quad (\chi \text{ ramified}), \qquad L(\chi) = \frac{1}{1-\chi(\pi)} \quad (\chi \text{ unramified}).$$

3.3. Given ψ and dx, put, for f a C^∞ function of compact support on K,

$$\widehat{f}(y) = \int_K f(x)\psi(xy)\,dx.$$

Define $\varepsilon(\chi, \psi, dx) \in \mathbb{C}^*$ by Tate's local functional equation, independent of f,

$$\frac{\int_{K^*} \widehat{f}(x)\,\omega_1\chi^{-1}(x)\,d^*x}{L(\omega_1\chi^{-1})} = \varepsilon(\chi, \psi, dx)\,\frac{\int_{K^*} f(x)\chi(x)\,d^*x}{L(\chi)}. \tag{3.3.1}$$

The same Haar measure d^*x is used on both sides. Both sides are defined by analytic continuation in χ in the families $\chi\omega_s$, $s \in \mathbb{C}$. In these families, ε is of the form Ae^{Bs}.

One has

$$\varepsilon(\chi, \psi, a\,dx) = a\,\varepsilon(\chi, \psi, dx), \tag{3.3.2}$$

$$\varepsilon(\chi, \psi(ax), dx) = \chi(a)\|a\|^{-1}\varepsilon(\chi, \psi, dx). \tag{3.3.3}$$

3.4. For K nonarchimedean, put:

$$n(\psi) = \text{the largest integer } n \text{ such that } \psi|_{\pi^{-n}\mathcal{O}} = 1;$$

$$a(\chi) = \begin{cases} 0, & \chi \text{ unramified},\\ \text{the least } m \text{ such that } \chi|_{1+(\pi)^m} = 1, & \chi \text{ ramified}; \end{cases}$$

$$sw(\chi) = \begin{cases} 0, & \chi \text{ unramified or tamely ramified},\\ a(\chi) - 1, & \chi \text{ ramified}; \end{cases}$$

and let γ be an element of K^* of valuation

$$v(\gamma) = n(\psi) + sw(\chi) + 1.$$

The local constant ε is given by (3.3.2), (3.3.3), and the following formulas.

$$K \simeq \mathbb{R}. \tag{3.4.1}$$

Let x be the embedding of K in $\mathbb{C}$, and $N = 0$ or 1. For $\psi = \exp(2\pi i x)$ and dx the Lebesgue measure,

$$\varepsilon(x^{-N}\omega_s, \psi, dx) = i^N.$$

$$K \simeq \mathbb{C}. \tag{3.4.2}$$

Let z be an embedding of K in $\mathbb{C}$, and $N \geq 0$. For

$$\psi = \exp(2\pi i\ \mathrm{Tr}_{\mathbb{C}/\mathbb{R}}(z))$$

and

$$dx = |dz \wedge d\overline{z}| = 2\,da\,db \quad (z = a + bi),$$

one has

$$\varepsilon(z^{-N}\omega_s, \psi, dx) = i^N.$$

3.4.3. K nonarchimedean. If χ is unramified,

$$\int_{\mathcal{O}} dx = 1, \qquad n(\psi) = 0,$$

then

$$\varepsilon(\chi, \psi, dx) = 1. \tag{3.4.3.1}$$

For χ ramified, one has

$$\varepsilon(\chi, \psi, dx) = \int_{K^*} \chi^{-1}(x)\psi(x)\, dx = \sum_n \int_{v(x)=n} \chi^{-1}(x)\psi(x)\, dx = \int_{\gamma^{-1}\mathcal{O}^*} \chi^{-1}(x)\psi(x)\, dx. \tag{3.4.3.2}$$

We shall see that it is natural to unify these two formulas as follows. If $\varepsilon_0(\chi, \psi, dx)$ is defined by

$$\varepsilon(\chi, \psi, dx) = \varepsilon_0(\chi, \psi, dx) \quad (\chi \text{ ramified}),$$

and

$$\varepsilon(\chi, \psi, dx) = \varepsilon_0(\chi, \psi, dx)\, \chi(\pi)^{-1} \quad (\chi \text{ unramified}), \tag{3.4.3.3}$$

then

$$\varepsilon_0(\chi, \psi, dx) = \int_{\gamma^{-1}\mathcal{O}^*} \chi^{-1}(x)\psi(x)\, dx. \tag{3.4.3.4}$$

From (3.4.3.3) and (3.4.3.4), one deduces that, for ω unramified,

$$\varepsilon_0(\chi\omega, \psi, dx) = \omega(\pi)^{n(\psi)+sw(\chi)+1} \varepsilon_0(\chi, \psi, dx), \tag{3.4.3.5}$$

and

$$\varepsilon(\chi\omega, \psi, dx) = \omega(\pi)^{n(\psi)+a(\chi)} \varepsilon(\chi, \psi, dx).$$

3.5. Suppose K nonarchimedean, and resume the notation of 2.2.1. Let V be a finite-dimensional vector space over a field E of characteristic 0, or over $\mathbb{C}$. The representations

$$\rho : W(\overline{K}/K) \longrightarrow \mathrm{GL}(V)$$

will always be assumed continuous: an open subgroup of I acts trivially. For V the space of a representation and V^I the subspace of inertia invariants, put

$$Z(V; t) = \det(1 - Ft^d, V^I)^{-1}, \tag{3.5.1}$$

$$L(V) = Z(V; 1) \qquad (E \in E^* \cup \{\infty\}). \tag{3.5.2}$$

Let

$$\omega^t : W(\overline{K}/K) \longrightarrow E((t))^*$$

be the unramified character such that $\omega^t(F) = t^d$. In an evident sense, $\omega_s = \omega^{p^{-s}}$. One has

$$Z(V, t) = L(V \otimes \omega^t). \tag{3.5.3}$$

3.6. It is so that (3.2.3) and (3.5.2) are compatible that we normalized the local class field theory isomorphism so that geometric Frobenius elements correspond to uniformizers. Both (3.2.3) and (3.5.2) are very natural: 3.2.3 is forced from the analytic point of view (cf. 3.3.1), and 3.5.2 follows the tradition of geometers; it is geometric Frobenius which tends to act on the cohomology of algebraic varieties with algebraic integers as eigenvalues.

3.7. The irreducible continuous complex representations of $W(\overline{K}/K)$ are, for K complex, the quasi-characters. For K real, besides the quasi-characters of K^*, they are the representations induced by quasi-characters χ of $\overline{K}^*$ such that $\chi \neq \chi \circ \sigma$, where σ is complex conjugation of $\overline{K}$.

For any complex representation V, expressed in the Grothendieck group of representations of $W(\overline{K}/K)$ as a sum of simple representations, define $L(V)$ as follows:

$$K \simeq \mathbb{R}, \quad V = \sum_i [\chi_i] + \sum_j \mathrm{Ind}_{\overline{K}^*}^{W}(\chi_j), \tag{3.7.1}$$

$$L(V) = \prod_i L_K(\chi_i) \prod_j L_{\overline{K}}(\chi_j).$$

$$K \simeq \mathbb{C}, \quad V = \sum_i [\chi_i], \qquad L(V) = \prod_i L(\chi_i). \tag{3.7.2}$$

Proposition 3.8. Let K be a local field and $\overline{K}$ an algebraic closure of K.

(i) For every exact sequence
$$0 \longrightarrow V' \longrightarrow V \longrightarrow V'' \longrightarrow 0$$
of complex representations of $W(\overline{K}/K)$, one has
$$L(V) = L(V')L(V''). \tag{3.8.1}$$

(ii) Let L be a finite separable extension of K in $\overline{K}$, and let V_L be a complex representation of $W(\overline{K}/L)$. Let V_K be the representation induced to $W(\overline{K}/K)$. Then
$$L(V_K) = L(V_L). \tag{3.8.2}$$

Assertion (i) is trivial. In the archimedean case, assertion (ii) follows from the duplication formula
$$\Gamma_{\mathbb{C}}(s) = \Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1),$$
corresponding to
$$\mathrm{Ind}([\omega_s]) = [\omega_s] + [\chi^{-1}\omega_{s+1}].$$

We prove (ii) for K nonarchimedean. Let k and ℓ be the residue fields of K and L. Then
$$V_K^{I_K} = \left(\mathrm{Ind}_{W(\overline{K}/L)}^{W(\overline{K}/K)} V_L\right)^{I_K} \simeq \mathrm{Ind}_{W(\overline{\ell}/\ell)}^{W(\overline{k}/k)}(V_L^{I_L})$$
as $W(\overline{k}/k)$-representations, both sides being identified with
$$\mathrm{Hom}_{W(\overline{K}/L)}(W(\overline{k}/k), V_L),$$
the covariant functions from $W(\overline{k}/k)$ to V_L, $W(\overline{K}/L)$ acting on $W(\overline{k}/k)$ by translation; cf. the commutative diagram

$$\begin{array}{ccc} W(\overline{K}/L) & \longrightarrow & W(\overline{\ell}/\ell) \\ \downarrow & & \downarrow \\ W(\overline{K}/K) & \longrightarrow & W(\overline{k}/k). \end{array}$$

This reduces (3.8.2) to the following well-known lemma.

Lemma 3.9. Let F be an endomorphism of V, let n be an integer, and let F_n be the endomorphism of

$$V_n = V \otimes \mathbb{C}^n$$

such that, for a basis $e_0, \ldots, e_{n-1}$ of $\mathbb{C}^n$,

$$F_n(v \otimes e_i) = v \otimes e_{i+1} \quad (0 \le i < n-1), \qquad F_n(v \otimes e_{n-1}) = F(v) \otimes e_0.$$

Then

$$\det(1 - F_n t) = \det(1 - F t^n).$$

By the principle of continuation of algebraic identities, one may suppose F diagonal in a suitable basis of V. One is immediately reduced to the case $\dim(V) = 1$, where F is multiplication by a number a^n, $a \neq 0$. In the basis $f_i = a^{-i}(v \otimes e_i)$, one has $F_n(f_i) = a f_{i+1}$, for $i \in \mathbb{Z}/n\mathbb{Z}$. The eigenvectors of F_n are the

$$\sum_i \chi(i) f_i$$

where χ runs through the characters of $\mathbb{Z}/n\mathbb{Z}$, and the determinant identity follows.

3.10. Let K be a global field, an $\mathfrak{A}$-field in the sense of [15]. For every place v of K, write K_v for the local field which is the completion of K at v. The objects defined above for local fields will be denoted, for K_v, with a subscript v. Let $\mathbb{A}$ be the adele ring of K, let $\| \ \|$ be the absolute value satisfying

$$d(ax) = \|a\| \, dx$$

for a Haar measure dx on $\mathbb{A}$, and let ω_s be the quasi-character $\|x\|^s$ of $\mathbb{A}^*$. If x_v is the component of $x \in \mathbb{A}$ in K_v, then

$$\|x\| = \prod_v \|x_v\|_v.$$

Let dx denote the unique Haar measure on $\mathbb{A}$ such that

$$\int_{\mathbb{A}/K} dx = 1$$

(the Tamagawa measure), let d^*x be a Haar measure on $\mathbb{A}^*$, and let ψ be a character trivial on $\mathbb{A}/K$, with local components ψ_v. For f a C^∞ function with compact support on $\mathbb{A}$, put

$$\widehat{f}(y) = \int_{\mathbb{A}} f(x)\psi(xy)\, dx.$$

3.11. Tate's global functional equation [14] says, for χ a quasi-character of $\mathbb{A}^*/K^*$,

$$\int_{\mathbb{A}^*} \widehat{f}(x)\chi'(x)\, d^*x = \int_{\mathbb{A}^*} f(x)\chi(x)\, d^*x, \qquad \chi' = \omega_1 \chi^{-1}, \tag{3.11.1}$$

the integrals being defined by analytic continuation in the families $\chi\omega_s$. Put

$$L(\chi) = \prod_v L(\chi_v) \tag{3.11.2}$$

by analytic continuation, and, for $dx = \otimes_v dx_v$,

$$\varepsilon(\chi) = \prod_v \varepsilon(\chi_v, \psi_v, dx_v). \tag{3.11.3}$$

The product (3.11.3) is independent of the choice of ψ and of the decomposition of dx, as follows from (3.3.2) and (3.3.3). If, for almost all v,

$$\int_{\mathcal{O}_v} dx_v = 1,$$

then almost all its factors are equal to 1. Formulas (3.3.1) and (3.11.1) give the global functional equation of Hecke L-functions:

$$\varepsilon(\chi)L(\chi') = L(\chi). \tag{3.11.4}$$

3.12. The results stated below are proved in [1] and [19]. Let K be a global field, $\overline{K}$ an algebraic closure of K, and V a complex representation of $\mathrm{Gal}(\overline{K}/K)$, always assumed continuous: the action of $\mathrm{Gal}(\overline{K}/K)$ factors through a finite Galois group $\mathrm{Gal}(K'/K)$. For every place v of K, choose a place of $\overline{K}$ above it, and let V_v be the representation obtained from V by restriction to the subgroup

$$W(\overline{K}_v/K_v)$$

of the decomposition group $\mathrm{Gal}(\overline{K}_v/K_v)$.

(A) For χ a quasi-character of $\mathbb{A}^*/K^*$, with local components χ_v, the infinite product

$$L(V,\chi) = \prod_v L(V_v \otimes \chi_v)$$

can be defined by analytic continuation in χ in the families $\chi\omega_s$. It converges for $\Re(s)$ large and extends to a meromorphic function of s.

(B) Let V^* denote the dual of V. One has

$$L(V,\chi) = \varepsilon(V,\chi)\, L(V^*,\chi'), \qquad \chi' = \omega_1\chi^{-1},$$

with $\varepsilon(V,\chi) \in \mathbb{C}^*$, of the form Ae^{Bs} as χ varies in the families $\chi\omega_s$.

(C) Let L be a finite extension of K in $\overline{K}$, let $N_{L/K}$ be the norm, let V_L be a representation of $\mathrm{Gal}(\overline{K}/L)$, and put

$$V_K = \mathrm{Ind}_{\mathrm{Gal}(\overline{K}/L)}^{\mathrm{Gal}(\overline{K}/K)}(V_L).$$

For χ a quasi-character of $\mathbb{A}_K^*/K^*$, one has

$$L(V_K,\chi) = L(V_L, \chi \circ N_{L/K}),$$

and similarly

$$\varepsilon(V_K,\chi) = \varepsilon(V_L, \chi \circ N_{L/K}).$$

(D) One also has

$$\varepsilon(V' \oplus V'', \chi) = \varepsilon(V',\chi)\varepsilon(V'',\chi),$$

which makes it possible to define $\varepsilon(V,\chi)$ when V is only a virtual representation.

The assertions (A) and (B) are proved by reduction to the case $\dim V = 1$, via Brauer's theorem. Assertion (C) follows from (B) and from the compatibility of norm and transfer recalled above.

4. Existence of local constants

Theorem 4.1. There exists a unique function ε, satisfying conditions (1) to (4) below, which associates a number

$$\varepsilon(V,\psi,dx) \in \mathbb{C}^*$$

to each isomorphism class of sextuples $(K,\overline{K},\psi,dx,V,\rho)$, consisting of a local field K, an algebraic closure $\overline{K}$ of K, a nontrivial additive character $\psi : K \to \mathbb{C}^*$, a Haar measure dx on K, a finite-dimensional complex vector space V, and a representation

$$\rho : W(\overline{K}/K) \longrightarrow \mathrm{GL}(V).$$

(1) For every exact sequence of representations

$$0 \to V' \to V \to V'' \to 0,$$

one has

$$\varepsilon(V,\psi,dx) = \varepsilon(V',\psi,dx)\varepsilon(V'',\psi,dx).$$

This condition shows that $\varepsilon(V,\psi,dx)$ depends only on the class of V in the Grothendieck group of representations of $W(\overline{K}/K)$, and allows $\varepsilon(V,\psi,dx)$ to be interpreted when V is only a virtual representation.

(2)
$$\varepsilon(V,\psi,a\,dx) = a^{\dim V}\varepsilon(V,\psi,dx).$$

In particular, for virtual V of dimension 0, $\varepsilon(V,\psi,dx)$ is independent of dx; it is denoted $\varepsilon(V,\psi)$.

(3) If L is a finite separable extension of K in $\overline{K}$, and V_K is the virtual representation of $W(\overline{K}/K)$ induced by a virtual representation of dimension zero V_L of $W(\overline{K}/L)$, then

$$\varepsilon(V_K,\psi) = \varepsilon(V_L,\psi\circ \mathrm{Tr}_{L/K}).$$

(4) If $\dim V = 1$, and ρ is defined by a quasi-character χ of K^*, then

$$\varepsilon(V,\psi,dx) = \varepsilon(\chi,\psi,dx).$$

4.2. Let
$$v = \sum_{i,j} n_{ij}(H_i,\chi_{ij}) \in R_0^+(W(\overline{K}/K))$$
(1.10). Assume that the H_i are pairwise nonconjugate. Let K_i be the extension of K defined by H_i, and again denote by χ_{ij} the quasi-character of K_i^* defined by χ_{ij}. For any choice of Haar measures dx_i on the K_i, and for ψ an additive character of K, put

$$\varepsilon(v,\psi) = \prod_{i,j}\varepsilon(\chi_{ij},\psi\circ\mathrm{Tr}_{K_i/K},dx_i)^{n_{ij}}. \tag{4.2.1}$$

It follows from (3.3.2) that $\varepsilon(v,\psi)$ is independent of the choice of the dx_i.

Lemma 4.3. With the notation 1.9.8, for $v\in R_0^+(W(\overline{K}/K))$,

$$\varepsilon(v,\psi(ax)) = \det(v)(a)\,\varepsilon(v,\psi).$$

It is enough to check this for $v=(H,\chi)-(H,1)$, where H corresponds to an extension L of K. Let v_0 be the virtual representation $[\chi]-[1]$ of $W(\overline{K}/L)$. Formula (3.3.3) gives the assertion over L. It is known ([10], XII, §4) that the inclusion of K^* into L^* is identified with the transfer

$$W(\overline{K}/K)^{\mathrm{ab}} \longrightarrow W(\overline{K}/L)^{\mathrm{ab}}.$$

Since $b(v)=\mathrm{Ind}(v_0)$, the lemma follows from 1.2.

Lemma 4.4. The theorem is true in the archimedean case.

For $K\simeq\mathbb{C}$, define ε by (1) and (4), and (2) is automatic. For $K\simeq\mathbb{R}$, one checks from 2.9 that

$$b: R^+(W(\overline{K}/K)) \longrightarrow R(W(\overline{K}/K))$$

is surjective, with kernel generated by the $\chi_s-\chi_t$ where

$$\chi_s = (K^*,\omega_s) - [\omega_s] - [\chi^{-1}\omega_{s+1}].$$

For a virtual representation V, and $v\in R_0^+(W(\overline{K}/K))$ such that

$$b(v) = V - \dim(V)[1],$$

put
$$\varepsilon(V,\psi,dx) = \varepsilon(v,\psi)\,\varepsilon(1,\psi,dx)^{\dim V}. \tag{4.4.1}$$

We check that the right-hand side is independent of the choice of v, i.e. that, for v satisfying $b(v)=0$, one has $\varepsilon(v,\psi)=1$. By 4.3, it is enough to check this for $\psi(x)=\exp(2\pi ix)$. For this choice of ψ, the assertion follows from the preceding determination of $\mathrm{Ker}(b)$, and from the fact that the right-hand sides of (3.4.1), (3.4.2) are independent of s. It is clear that ε defined by (4.4.1) satisfies (1) to (4).

We now restrict to the nonarchimedean case.

4.5. Let K be a nonarchimedean local field, $\overline{K}$ a separable closure of K, and $K_{nr} \subset \overline{K}$ the maximal unramified extension of K. Let J be an open normal subgroup of the inertia group

$$I = \mathrm{Gal}(\overline{K}/K_{nr})$$

and let L be the corresponding extension of K_{nr}. For $\sigma \in I/J$, put

$$t_{I/J}(\sigma) = \inf\bigl(v_L(\sigma(x) - x) \mid x \text{ integral in } L\bigr),$$

and define functions $a_{I/J}$ and $sw_{I/J}$ by

$$\begin{aligned} a_{I/J}(\sigma) &= -t_{I/J}(\sigma) \quad (\sigma \neq e), \\ sw_{I/J}(\sigma) &= 1 - t_{I/J}(\sigma) \quad (\sigma \neq e), \\ \sum_\sigma a_{I/J}(\sigma) &= \sum_\sigma sw_{I/J}(\sigma) = 0. \end{aligned} \tag{4.5.1}$$

By Artin (see [10], VI, §2, th. 1), $a_{I/J}$ is the character of a representation $A_{I/J}$ of I/J, the Artin representation. By loc. cit., prop. 2, $sw_{I/J}$ is likewise the character of a representation $Sw_{I/J}$, the Swan representation. These representations are defined only up to isomorphism. $A_{I/J}$ is the sum of $Sw_{I/J}$ and the augmentation representation of I/J. For $I \supset J \supset K$, one has (loc. cit., prop. 3)

$$(A_{I/K})^{J/K} \xrightarrow{\sim} A_{I/J}, \qquad (Sw_{I/K})^{J/K} \xrightarrow{\sim} Sw_{I/J}. \tag{4.5.2}$$

If d is the valuation of the discriminant $\delta_{L/K_{nr}}$, i.e. the valuation of the different $\mathfrak{D}_{L/K}$ ([10], III, §3, prop. 6), and r denotes a regular representation, then (loc. cit., prop. 4)

$$A_{I/K}|_{J/K} = d\, r_{J/K} + A_{J/K}. \tag{4.5.3}$$

Let V be a representation of $W(\overline{K}/K)$, trivial on $J \subset I$, with character ψ. Define the Artin and Swan conductors of V by

$$\begin{aligned} a(V) &= \dim \mathrm{Hom}_{I/J}(A_{I/J}, V) = sw(V) + \dim V - \dim V^I, \\ sw(V) &= \dim \mathrm{Hom}_{I/J}(Sw_{I/J}, V) = \frac{1}{|I/J|} \sum_\sigma sw_{I/J}(\sigma)\psi(\sigma). \end{aligned} \tag{4.5.4}$$

By 4.5.2, these conductors are independent of the choice of J. For V of dimension 1, defined by a quasi-character χ, one has (loc. cit., prop. 5)

$$a(V) = \text{the conductor } a(\chi) \text{ of } \chi. \tag{4.5.5}$$

The preceding notation is therefore compatible with that introduced in 3.4.

With the notation 1.9.8, one has:

Lemma 4.5. For $v \in R_0^+(W(\overline{K}/K))$ and χ an unramified quasi-character of K^*,

$$\varepsilon(v \cdot \chi, \psi) = \chi(\pi)^{a(v)} \varepsilon(v, \psi).$$

It suffices to verify this for $v = (H, \lambda) - (H, \mu)$, where H defines an extension L of K, and λ, μ are two quasi-characters of L^*. Let d be the valuation of the different of L/K, let f be the residue-degree, let $I \subset W(\overline{K}/K)$ and $J \subset W(\overline{K}/L)$ be the inertia groups, and choose $K \subset J$ small enough. We compute $a(v)$. The adjunction formula for induced representations, (4.5.3), and (4.5.5) give, for every quasi-character ν of H, i.e. of L^*,

$$a((H,\nu)) = \dim \mathrm{Hom}_{I/K}(A_{I/K}, \mathrm{Ind}_H^W(\nu)) = f\, \dim \mathrm{Hom}_{J/K}(A_{J/K} + d\, r_{J/K}, [\nu]|_J) = f\, a(\nu) + fd. \tag{4.5.1'}$$

Thus

$$a(v) = f(a(\lambda) - a(\mu)).$$

For ν a quasi-character of L^*, χ_1 an unramified quasi-character of L^*, and $n = n(\psi \circ \mathrm{Tr}_{L/K})$ as in 3.4, one has by (3.4.3.1) or (3.4.3.2)

$$\frac{\varepsilon(\nu\chi_1, \psi \circ \mathrm{Tr}, dx)}{\varepsilon(\nu, \psi \circ \mathrm{Tr}, dx)} = \chi_1(\pi_L)^{a(\nu)+n}.$$

Hence

$$\varepsilon(v \cdot \chi, \psi)\varepsilon(v, \psi)^{-1} = \chi \circ N_{L/K}\bigl(\pi_L^{a(\lambda)-a(\mu)}\bigr) = \chi(\pi)^{f(a(\lambda)-a(\mu))},$$

which is the announced formula.

Terminology 4.6. Let K_1 be a Galois extension, finite or not, of K, with Galois group G. Let α be a quasi-character of K^*, and ψ a nontrivial additive character of K. We shall say that there exists an α, ψ-theory of constants for K_1/K if there exists $\varepsilon_{\alpha,\psi}$ of the following type:

(0) For H an open subgroup of G, defining a finite extension L of K, for V a representation of H, and dx a Haar measure on L,

$$\varepsilon_{\alpha,\psi}(V, dx) \in \mathbb{C}^*$$

is defined.

(1) For every exact sequence of representations

$$0 \to V' \to V \to V'' \to 0,$$

one has

$$\varepsilon_{\alpha,\psi}(V, dx) = \varepsilon_{\alpha,\psi}(V', dx)\varepsilon_{\alpha,\psi}(V'', dx).$$

Thus $\varepsilon_{\alpha,\psi}$ still makes sense for virtual representations.

(2) One has

$$\varepsilon_{\alpha,\psi}(V, a\,dx) = a^{\dim V}\varepsilon_{\alpha,\psi}(V, dx).$$

In particular, for virtual V of dimension 0, $\varepsilon_{\alpha,\psi}(V, dx)$ is independent of dx; it is denoted $\varepsilon_{\alpha,\psi}(V)$.

(3) For $H_1 \subset H_2 \subset G$, defining $L_1 \supset L_2$, and V a virtual representation of dimension 0 of H_1, one has

$$\varepsilon_{\alpha,\psi}(\operatorname{Ind}_{H_1}^{H_2} V) = \varepsilon_{\alpha,\psi}(V).$$

(4) For V of dimension 1, defined by a character χ of L^*,

$$\varepsilon_{\alpha,\psi}(V, dx) = \varepsilon(\chi\,\alpha \circ N_{L/K},\, \psi \circ \operatorname{Tr}_{L/K},\, dx).$$

Lemma 4.7. Let K_1/K and α be as above, and let ψ be a nontrivial additive character of K. There exists at most one α, ψ-theory of constants for K_1/K. For one to exist, it is necessary and sufficient that, for every

$$v \in \operatorname{Ker}(R_0^+(G) \longrightarrow R(G)),$$

one have

$$\varepsilon(v \cdot \alpha, \psi) = 1.$$

Let $H \subset G$, L, and V be as above. By 1.5, there exists $v \in R_0^+(H)$ whose image in $R(H)$ is $V - \dim(V)[1]$. Necessarily

$$\varepsilon_{\alpha,\psi}(V, dx) = \varepsilon(v \cdot (\alpha \circ N_{L/K}),\, \psi \circ \operatorname{Tr}_{L/K})\, \varepsilon(\alpha \circ N_{L/K},\, \psi \circ \operatorname{Tr}_{L/K},\, dx)^{\dim V}, \tag{4.7.1}$$

which gives uniqueness. The hypothesis ensures that $\varepsilon_{\alpha,\psi}$, defined by this formula, is independent of the choice of v; one easily checks axioms (0) to (4).